Cyberpsychology, Behavior, and Social Networking
Volume 00, Number 00, 2025
© Mary Ann Liebert, Inc.
DOI: 10.1089/cyber.2024.0518

Supporting Brainstorming with a Collaborative Platform or a Generative Artificial Intelligence Tool: An Exploratory Study

Katusha Gerardini, MSc,[1] Eleonora Diletta Sarcinella, MSc,[1] Francesca Borghesi, MSc,[2] Andrea Pozzi, PhD,[3] Andrea Gaggioli, PhD,[1,4] and Alice Chirico, PhD[1]

Abstract

This study explores the potential of generative artificial intelligence (GAI) versus traditional whiteboards in supporting group brainstorming. Sixteen groups of five members each used Miro (a conventional whiteboard) and DALL-E (an image-based GAI tool), either online or offline, in a 2×2 experimental design (online vs. offline; Miro vs. DALL-E). Researchers measured participants' affect (Positive and Negative Affect Schedule), emotions (Aesthetic Emotions Questionnaire), creative self-efficacy, technological readiness (Technology Readiness Survey), user experience (User Experience Questionnaire), flow (Flow State Scale–Short Version), and creativity (fluidity, elaboration, flexibility, and originality). Two independent raters evaluated the groups' ideas for each member. Results showed that DALL-E generated more positive affect, richer esthetic experiences, and higher attractiveness and novelty than Miro, particularly online. A significant interaction effect was found for "efficiency" (UX dimension) and "union action-consciousness" (flow dimension). Participants felt more creative and preferred working with DALL-E. Moreover, online sessions with DALL-E led to greater idea elaboration. These findings suggest that GAI tools such as DALL-E could reshape and enhance traditional group creativity methods, making them core assets in group collaboration, especially in online settings.

Keywords: Creativity, Generative Artificial Intelligence, Brainstorming, Human–AI Collaboration

Downloaded by 37.159.0.213 from www.liebertpub.com at 06/18/25. For personal use only.

Introduction

Defining creativity with a single definition is nearly impossible, as it has been studied from multiple perspectives over time. Initially, creativity was seen as a reflection of divine inspiration,[1] with no recognition of individual contribution.[2] During the Renaissance, however, it became associated with the solitary creative genius,[3–5] an inspired figure linked to free imagination.[6] In the 20th century, Guilford[7] expanded this view, introducing a psychological and social dimension beyond the purely individual perspective.

Creativity has been analyzed not only as a distinctive quality of exceptional people but also as a capability accessible to all.[8] Models, such as divergent thinking[7,9] and integrative approaches, framed creativity as the result of interactions between individuals, environment, and culture.[10] Over time, this perspective has evolved into a view of creativity as a complex system,[11] emphasizing "distributed creativity,"[12] where it emerges from networks of people, cultural artifacts, materials, and their interaction. Therefore, group creativity emerges as a collective process built on individual contributions continuously influenced by peer input and shared environments. This shift laid the foundation for today's multidisciplinary understanding of creativity, where social context and digital technologies play a key role in idea production.[13]

These way emerging technologies such as artificial intelligence (AI) are redefining creative processes. AI has recently accelerated innovation across fields such as art,[14] music,[15,16] business,[17] and creative domains.[18,19]

Creativity often seen as an intrinsic human trait,[20] a human strength,[21–23] and cognitive ability[24–26] to generate original, useful, and influential ideas,[27] can be now impacted by generative artificial intelligence (GAI), which can rapidly produce multiple ideas, mimicking humanlike creativity,[18,28] especially by remixing existing content.[30,31] This evolution has sparked debate about GAI's potential to go beyond ideation and be considered fully creative. Studies comparing human and AI performance[19,33] often emphasize competition, using Guilford and Torrance's creativity dimensions: fluidity, flexibility, originality, and elaboration.[7]

[1]Research Center in Communication Psychology (PsiCom), Department of Psychology, Catholic University of the Sacred Heart, Milan, Italy.
[2]Department of Psychology, University of Turin, Turin, Italy.
[3]Faculty of Mathematical, Physical and Natural Science, Catholic University of the Sacred Heart, Brescia, Italy.
[4]IRCCS Istituto Auxologico Italiano, Milan, Italy.

Downloaded by 37.159.0.213 from www.liebertpub.com at 06/18/25. For personal use only.

Joosten et al.[18] found that in corporate settings, ChatGPT outperformed humans in novelty, customer benefit, and quality. However, Lavrič and Škraba[28] observed that human ideas had greater diversity (entropy), while AI was faster. Despite GAI's advantages, its role as a competitor or complement remains debated.

While solitude supports creativity,[34,35] group dynamics also play a role. Osborn[36] promoted brainstorming as a technique for generating innovative ideas, and subsequent studies examined how group factors stimulate creativity.[37] How GAI affects such dynamics as interaction, engagement, and output is still unclear. Balancing tradition and innovation means viewing GAI not just as a tool but as a collaborator. Our study investigates this integration and how GAI might enhance human creativity.

The rise of large language models (LLMs) has enhanced human–AI collaboration in creative activities,[38] often through prompting techniques.[39] This raises new questions on co-creativity.[40,41] Some researchers label AI output as "pseudo-creativity,"[42,43] while others[44] argue for a clear distinction between human and artificial creativity.

Emerging perspectives[45] predicted four possible scenarios of human–AI collaboration: (a) a "collaborative" scenario, where human and AI contributions would be equally valued, creating a hybrid context; (b) an "organic" scenario, where human-made creations gain greater authenticity and value; (c) a "Plagiarism 3.0" scenario, where individuals can rely solely on AI content without attribution; and (d) a "shutdown" scenario, where AI's perceived superiority discourages human creativity.

Our study explores human–AI collaboration, examining the emergence of a hybrid creative context.[46] GAI tools such as ChatGPT can enhance creative output in idea generation,[47] improving originality and solution quality in creative problem solving.[48] While brainstorming is a classical group creativity method,[36] digital versions such as brainwriting and electronic brainstorming now incorporate new tools. Research suggested that in-person brainstorming fosters spontaneous interaction, while online brainstorming may reduce social pressure but increase cognitive dispersion.[49]

GAI is now used in brainstorming,[50] aiding ideation despite limits such as dependence on historical data.[51] Still, it can support creativity,[52,53] as a partner.[54,55] However, group collaboration is complex, and GAI's role in managing cognitive load remains uncertain. Overreliance may lead to reduced individual input, the "free-riding" effect[56] as some rely too much on GAI,[57] lowering group performance.[58]

Since ChatGPT only provides text, it may increase effort during ideation. However, visualizing ideas has been shown to stimulate creativity and divergent thinking.[59] In traditional brainstorming, participants document all generated ideas during the divergent phase, aiding selection in the convergence phase while managing cognitive complexity.

Visual digital tools such as Miro[1] support idea generation, with digital post-its, providing effective support for creativity and feedback.[60] DALL-E[2], an image-based GAI, lets users quickly externalize mental images, vividly enhancing the creative process. Addressing the gap in research on visual and digital GAI tools in creative collaboration, this study explores how hybrid contexts can enhance idea generation and user experience. Specifically, this study explores the role of GAI in group brainstorming by comparing DALL-E with Miro, across online and in-person settings. It examines (a) What is GAI's role during online and offline brainstorming sessions? (b) How do the UX, flow, esthetic emotions, and affect change using DALL-E and Miro? (c) How do these tools affect the creative output, in fluidity, flexibility, elaboration, and originality?

Materials and Methods

This study followed the following experimental design with "*condition*" as a between-subjects variable (i.e., online vs. in-presence) and "*platform*" as a within-subjects variable (i.e., DALL-E vs. Miro) (see Figure 1). All methods were carried out following the Helsinki Declaration, and the Ethical Commission of the Catholic University of the Sacred Heart of Milan approved the study.

Sample

Eighty participants (45 females and 35 males) were involved, with a mean age of 33.8 (SD = 10.4). Most reported a master's degree. The sample's Technological Readiness Index was 14.1. Specifically, the mean for the optimism scale was 4.33 (SD = 0.82), for the innovation scale was 3.52 (SD = 1.15), while for technological discomfort was 2.70 (SD = 0.88), and for technological insecurity scale was 3.53 (SD = 0.98).

Seventy-three participants had never used DALL-E and Miro before the experiment, while 7 had. Specifically, three participants had already used Miro, but not DALL-E, while another three had used the opposite.

Recruitment

Participants were recruited via snowball sampling, starting from the researchers' contacts. Invitations were sent through personal networks, email, and social media. To avoid dependency relationships, a researcher selected each participant they had no prior acquaintance.

Inclusion criteria

- >18 years old.
- Native Italian speakers.
- High skills in using social online platforms such as Microsoft Teams.

Measures and instruments

The instruments were the following:

- Sociodemographic questions.
- Technology Readiness Survey (TRI 2.0)[61] measures people's readiness to adopt new technologies on four scales (i.e., Optimism, Innovation, Discomfort, and Insecurity).
- Positive and Negative Affect Schedule (PANAS),[62] which measures the intensity of positive and negative affective states.

[1]https://miro.com/it/.

[2]https://chatgpt.com.

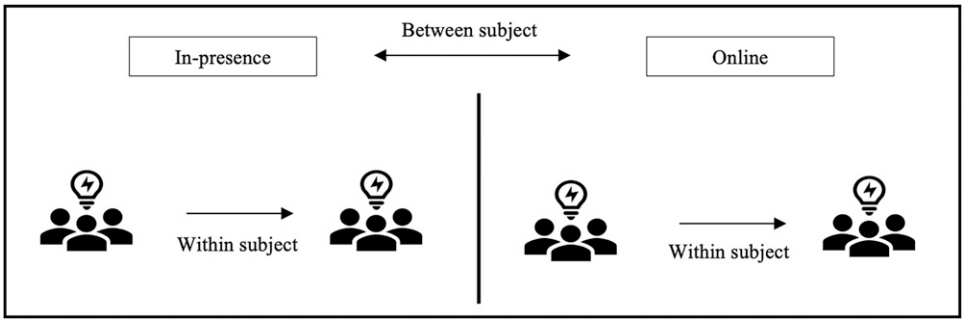


FIG. 1. Graphical representation of the experiment.

- Flow State Scale–Short Version (FSS),[63] which measures the state of flow for the experience with the nine scales (i.e., Balance Challenges-Skills, Union Action-Consciousness, Clear Goals, Immediate Feedback, Focus on Task, Sense of Control, Loss of Self-Consciousness, Transformation of Time, and Autotelic Experience).
- User Experience Questionnaire (UEQ),[64] to evaluate the overall users' experience according to six dimensions (i.e., attractiveness, perspicuity, efficiency, dependability, stimulation, and novelty).
- Aesthetic Emotions Questionnaire (AESTHEMOS),[65] which measures esthetic emotions with 21 subscales (i.e., feeling of beauty/liking, fascination, being moved, awe, enchantment, nostalgia, joy, humor, vitality, energy, relaxation, surprise, interest, intellectual challenge, insight, feeling of ugliness, boredom, confusion, anger, uneasiness, and sadness).
- Post-Experience Questionnaire with *ad hoc* questions measuring participants' preferred platform (i.e., what did you prefer: DALL-E vs. Miro? And why?) and their creative self-efficacy (i.e., did you feel more creative in Miro vs. DALL-E?).
- Brainstorming technique, to stimulate and then analyze creativity, applying a typical scoring of divergent thinking tests.[7]

Procedure

Participants completed a pre-experimental online survey including informed consent, sociodemographic questions, and TRI 2.0.

Then, participants were randomly assigned either to the in-presence or the online condition. In both cases, they were invited to take part in groups of five members in two different 15-minute brainstorming sessions, following a counterbalanced order, using two different instruments (a) Miro or (b) DALL-E (see Figures 2 and 3). They were asked to generate as many ideas as possible about either (a) the future airplane or (b) the future car, again in counterbalanced order. Participants were randomly assigned to groups to minimize bias from prior knowledge or relationships. They used Microsoft Teams online for real-time idea sharing and commenting, with all participants viewing the tools on their screens. In the in-presence condition, the setup was similar but without Microsoft Teams; instead, a shared screen allowed interaction

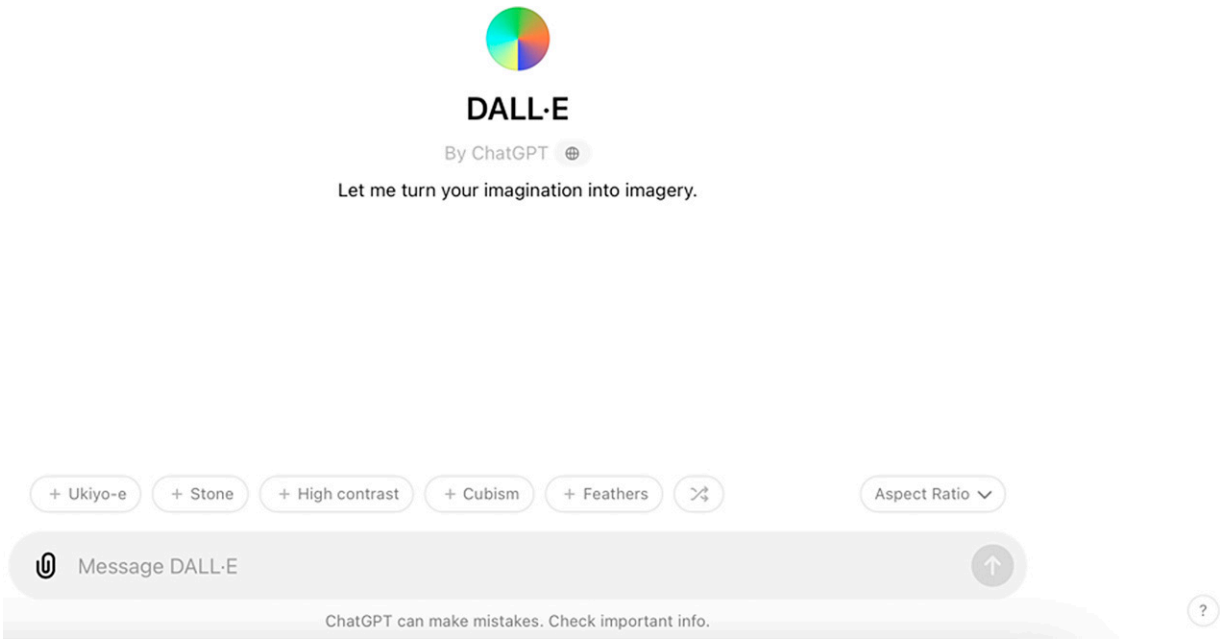


FIG. 2. A screenshot of DALL-E dashboard.

Downloaded by 37.159.0.213 from www.liebertpub.com at 06/18/25. For personal use only.

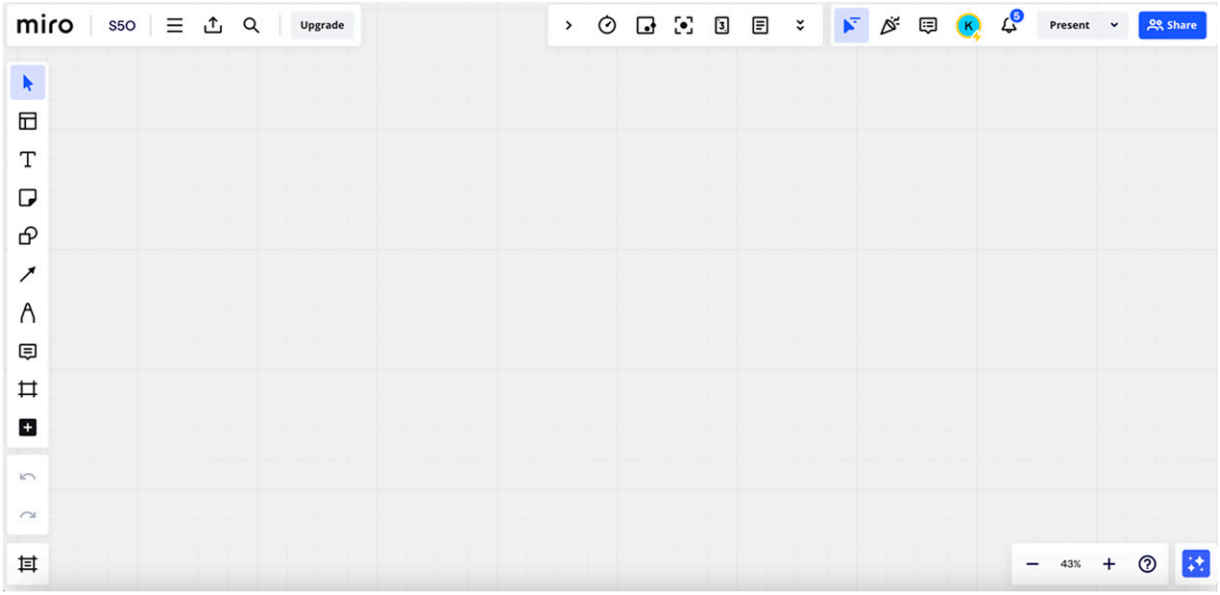


FIG. 3. A screenshot of Miro dashboard.

with the tools. Instructions were given to the participants before each session. Two tutorials were given on how to use the tools, regardless of whether they had used them before or not, and the basic rules of brainstorming. The decision to limit each brainstorming session to 15 minutes was based on previous research,[66,67] showing that limited brainstorming sessions are more likely to stimulate creativity and minimize mental fatigue and cognitive load. At the end of each session, participants completed the PANAS, UEQ, AESTHEMOS, and FSS. Finally, at the end of both brainstorming sessions, they were asked to indicate their favorite condition and the one in which they perceived themselves to be highly creative (*ad hoc* questions), also explaining the reasons for their preferences. All sessions were audio-recorded for later analysis of creativity by means of two independent raters.

Scoring criteria

The idea scoring was carried out by two independent raters who were experts in Guilford's scoring criteria, which were the following:

- Fluidity: total number of ideas generated by each participant.
- Elaboration: number of details for each idea. Here, we computed the means of elaboration for each participant.
- Flexibility: number of semantic categories.
- Originality: frequency of occurrence of each idea out of the total number of unique ideas.

Raters resolved disagreements through comparison; if unresolved, a third independent expert made the final decision.

Data Analyses

First, we computed the descriptives to test whether variables approximated a normal distribution, checking for skewness and kurtosis. At the individual level, we computed mixed analyses of variance (ANOVAs) with "platform" and "condition" as independent variables, with the following measures: Balance Challenges-Skills, Union Action-Consciousness, Clear Goals, Immediate Feedback, Concentration, Sense of Control, Loss of Self-awareness, Deconstruction of Time and Autotelic Experience, Attractiveness, Perspicuity, Efficiency, Dependability, Stimulation and Novelty, Feeling of Beauty/Liking, Fascination, Being Moved, Awe, Enchantment, Nostalgia, Joy, Humor, Vitality, Energy, Relaxation, Surprise, Interest, Intellectual Challenge, Insight, Feeling of Ugliness, Boredom, Confusion, Anger, Uneasiness, and Sadness.

First, we discussed the interaction effects and then the main ones. Second, we compared the creativity scores (fluidity, elaboration, flexibility, and originality) for each instrument, Miro and DALL-E, in both online and in-presence conditions, again at the individual level.

To establish interrater reliability for the creativity scores, we computed Pearson's r coefficients between ratings by the two independent judges for each of the four dimensions (fluency, flexibility, elaboration, and originality). Then, since the correlations were medium to high ($r = 0.50$–0.90), indicating satisfactory raters' agreement, we calculated the mean score of each dimension across the two raters to get a more stable and representative measure. Furthermore, we computed four mixed ANOVAs for the creativity dimensions. Finally, we computed the frequency of response for preferred experience and creative self-efficacy.

Results

Regarding user experience dimensions, between-subjects variable "condition" (online vs. offline), a significant main effect of mode of performance on novelty was found, $F(1) = 3.89$; $p = 0.052$; $\eta^2 p = 0.047$. Furthermore, a significant interaction effect was found between "condition" (online vs. offline) × "platform" (DALL-E vs. Miro) for Efficiency, $F(1) = 5.507$; $p = 0.021$; $\eta^2_p = 0.066$ (see Figure 4).

Figure 4 shows how the average efficiency varies between platforms (Miro and DALL-E) and conditions (online and

Downloaded by 37.159.0.213 from www.liebertpub.com at 06/18/25. For personal use only.

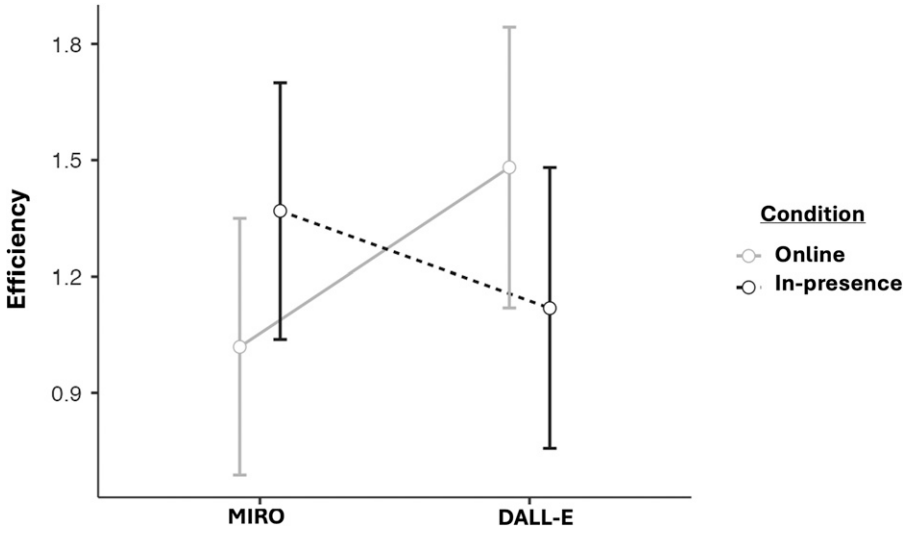


FIG. 4. Interaction graph between platform (Miro and DALL-E) and condition (online and in-presence) with efficiency as a measure.

Downloaded by 37.159.0.213 from www.liebertpub.com at 06/18/25. For personal use only.

in-presence). The results showed that the efficiency of participants is influenced by the interaction between the platform used and the condition in which the task was performed. In particular, the data suggest that DALL-E is most effective when used online, while Miro is more effective when used in-presence.

Concerning flow, a significant interaction effect was found between "condition" (online vs. offline) × "platform" (DALL-E vs. MiroRO) for Union Action-Consciousness variables, $F(1) = 5.35$; $p = 0.023$; $\eta^2_p = .064$ (see Figure 5).

Figure 5 shows how the average action-consciousness union varies between platforms (Miro and DALL-E) and conditions (online and in-person). The results indicated that the action-consciousness union is influenced by the interaction between the platform and the task conditions. Specifically, DALL-E provides a higher action-consciousness union online, while Miro performs better in the in-presence condition.

On an individual level, 83% of participants preferred the DALL-E experience, and 76% felt more creative with DALL-E.

Regarding creativity, a significant main effect of "condition" (online vs. offline) on elaboration's creativity dimensions was found, $F(1) = 4.12$; $p = 0.046$; $\eta^2_p = 0.050$.

Discussion

At the affective level, this study showed that brainstorming with DALL-E elicited significantly more positive affect than Miro. DALL-E experience significantly increased the intensity of the following esthetic dimensions: Beauty, Fascination, Being Moved, Awe, Enchantment, Joy, Humor, Vitality, Energy, Surprise, Interest, Intellectual Challenge, and Insight (see Table 1). In line with this, and conversely, brainstorming with Miro conveyed a significantly higher feeling of ugliness than with DALL-E. It had been shown that brainstorming with DALL-E evoked not just a more intense basic positive affect but especially more intense esthetic positive emotions than Miro. These results may suggest a potential capacity of the tool to enrich the overall quality of user interaction.[68] On the contrary, they could be influenced by the novelty of

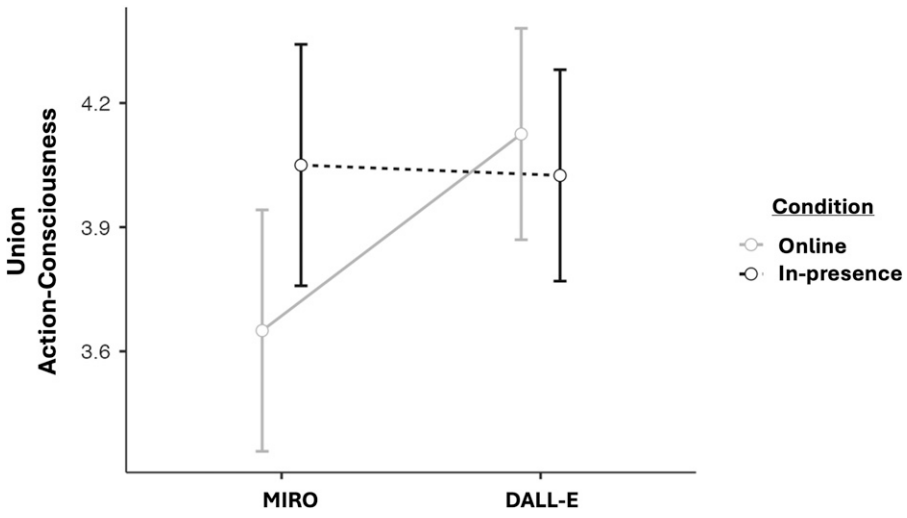


FIG. 5. Interaction graph between platform (Miro and DALL-E) and condition (online and in-presence) with union action-consciousness as a measure.

Table 1. Summary of Significant Main Effects of Mixed Analyses of Variance, Within-Subjects Variables "Platform"

	ANOVA			
Variable	F	*df*	p	$\eta^2 p$
Positive Affect	20.38	1	<0.001	0.21
Union Action-Consciousness	4.33	1	<0.04	0.053
Transformation of Time	6.32	1	<0.01	0.08
Autotelic Experience	15.66	1	<0.001	0.17
Attractiveness	15.31	1	<0.001	0.19
Stimulation	23.79	1	<0.001	0.23
Novelty	29.45	1	<0.001	0.27
Feeling of Beauty	21.76	1	<0.001	0.32
Fascination	43.03	1	<0.001	0.35
Being Moved	20.68	1	<0.001	0.210
Awe	41.82	1	<0.001	0.35
Enchantment	40.28	1	<0.001	0.34
Joy	30.09	1	<0.001	0.28
Humor	33.65	1	<0.001	0.30

ANOVA, analysis of variance.

DALL-E, though this was methodologically controlled by counterbalancing the conditions. Novelty effects can temporarily enhance user experiences, especially when interacting with innovative or visually stimulating technologies. The unfamiliarity with DALL-E may have amplified users' affective and esthetic responses, potentially influencing evaluations of stimulation and attractiveness. However, this effect was mitigated by providing participants with a brief tutorial on each tool before the brainstorming sessions.

Theoretical models of group creativity highlighted the critical role of the environment in shaping creative processes. Digital tools and artifacts used in brainstorming can either enhance or hinder creativity by influencing cognitive load and focus. DALL-E, offering vivid visual feedback, may accelerate creative associations, with research showing that exposure to diverse perspectives and unexpected stimuli can disrupt habitual thinking and deepen idea exploration.[69]

Indeed, a significant interaction effect emerged about efficiency, as a UX dimension. In online settings, DALL-E was perceived as significantly more efficient than Miro, whereas Miro outperformed DALL-E in in-person sessions. Moreover, brainstorming with DALL-E was also perceived as more attractive, stimulating, and novel than brainstorming with Miro. These findings were in line with the recent introduction of DALL-E, as a new and not yet fully explored instrument, while Miro has been in use since 2011. Participants also favored DALL-E, possibly due to its novelty and complexity, which enhances arousal and curiosity[70]—a hypothesis supported by self-reported data. Previous studies suggest that perceived visual attractiveness can impact a tool's usefulness, enjoyment, and ease of use,[71] which may have been a factor here.

Effective group brainstorming relies on balancing divergent and convergent thinking.[72] DALL-E's ability to produce vivid, unexpected visuals likely stimulated creative exploration, while Miro's structured, text-based format aided idea refinement during convergence. This interplay between exploration and refinement is a key element of successful group creativity.

Miro was perceived as less novel and surprising than DALL-E. Moreover, considering that positive affect has often been associated with an increase in divergent thinking[73–75] and in flow (e.g.,[76]), it was not surprising to find that brainstorming with DALL-E was also associated with a more intense experience of flow given its inherently emotionally positive nature. DALL-E induced a significantly higher sense of Union of Action-Consciousness, Deconstruction of Time, and Autotelic Experience, as dimensions of flow. Specifically, participants felt more blended with the task (Union Action-Consciousness) in the online brainstorming with DALL-E and in the in-presence brainstorming with Miro. Overall, DALL-E tended to elicit stronger task immersion during brainstorming sessions.

This aligns with Zhang et al.[77] who suggested that intrinsic motivation enhances cognitive effort in online creativity rather than direct group interaction. This may explain why DALL-E's novelty and surprise fostered flow and engagement: its emotionally stimulating nature likely sustained participants' motivation. The unpredictability of visual outcomes triggered curiosity and persistence, key elements of intrinsic motivation, further increasing engagement.

The DALL-E experience was perceived as more intrinsically motivating and pleasurable, likely due to the novel skills required for its use. Regarding perceived novelty, a main effect emerged favoring DALL-E, as well as a difference between online and in-person conditions, though without a significant interaction. This may be explained by the unique engagement of DALL-E: its novelty and the unpredictability of outcomes made the experience more surprising. The images generated by the AI vividly reflected participants' inner ideas, functioning like a "mental projector"—a quality not matched by Miro, as supported by participants' qualitative reports. The main effect of the condition on novelty may also stem from the increasing familiarity with online collaboration platforms such as Microsoft Teams, which have become widespread since the 2020 pandemic. The strong preference for DALL-E and participants' reports of feeling that it accurately visualized their ideas highlight a perceived alignment between user intention and AI output. This may boost engagement and flow but also blur the line between human input and machine suggestion, raising concerns about creative authenticity. In group contexts, this challenges authorship and ownership, making individual contributions harder to trace. It also raises ethical questions about agency and overreliance on GAI. Future studies should explore how perceptions of "true authenticity"[78,79] shape engagement and creative self-efficacy.

Regarding creativity, participants produced significantly more detailed ideas with Miro than DALL-E (see Table 2 and Figure 6). Online sessions, regardless of tool, led to more detailed ideas than in-person ones. Using GAI in brainstorming may encourage free-riding behaviors, especially online and among adults. Interestingly, this contrasts with the greater elaboration seen in online sessions, where ideas were more detailed and refined. This paradox suggests that while online environments may foster some disengagement, they also provide conditions, such as asynchronous interactions and reduced social pressure, that enhance idea depth and complexity. While some participants may have reduced their effort, others may have felt freer to explore and expand

Downloaded by 37.159.0.213 from www.liebertpub.com at 06/18/25. For personal use only.

Table 2. Descriptive Statistics of Creativity's Dimensions

		Miro		DALL-E	
Variable	*Condition*	*M*	*SD*	*M*	*SD*
Fluidity	Online	5.83	1.77	5.35	1.87
	In-presence	6.45	2.06	6.03	2.33
Elaboration	Online	1.79	1.76	2.23	1.47
	In-presence	1.68	.81	1.69	1.72
Flexibility	Online	4.53	1.47	4.22	1.62
	In-presence	4.33	1.72	4.17	1.77
Originality	Online	7.14	4.97	7.94	3.02
	In-presence	7.05	3.29	7.15	5.65

M, mean; SD, standard deviation.

ideas without social pressure. These differences may stem from the fact that group ideation relies not only on participants but also on artifacts[80] and the surrounding environment.[81] In this case, online brainstorming served as both an impactful artifact and environment, highlighting the importance of brainstorming modality. This might be due to the potential of online platforms to enhance creativity by overcoming creative blocks,[82] although online brainstorming could also risk decreasing engagement and satisfaction, compared with offline brainstorming.[83] Furthermore, Miro's lack of strong visual stimulation may have redirected participants' cognitive resources toward deepening ideas rather than generating novel but less developed outputs. However, it is important to note that free-riding was not systematically measured in this study, and observations are based on qualitative impressions. Further research is needed to explore how combining these tools could balance novelty-driven exploration with detailed elaboration and the free-riding phenomenon.

These findings reflect a broader shift from competitive to collaborative perspectives on human–AI creativity. Rather than viewing GAI as a competitor, participants appeared to engage with it as a cognitive partner that supported idea development. The preference for DALL-E and higher flow suggests an active human–AI co-construction process. This aligns with recent frameworks[46] where creativity emerges from the interplay between human cognition, social dynamics, and digital tools, with GAI contributing as an active agent in the creative flow.

Though some effects (e.g., $\eta^2 p < 0.07$) were modest, the differences still indicate meaningful patterns, even if their practical impact is limited. Still, the study highlights the potential of image-based GAI like DALL-E in education and business. In creative disciplines, GAI can help students visualize ideas, increase engagement, and build confidence (e.g., architecture students visualizing concepts before refining them). In business, it can support early-stage projects, particularly in product design and advertising, by offering rapid visual input and helping teams overcome creative blocks. As a "creativity catalyst" in remote collaboration, GAI can stimulate divergent thinking without replacing human input, enabling facilitators to enrich group ideation and foster dynamic, engaging environments. GAI could also benefit health care and well-being. In art therapy, for example, DALL-E could transform children's verbal expressions into pictures, helping therapists to stimulate dialogue and explore emotions.

Limitations and future directions

This exploratory study has several limitations that inform future research. First, only two tools, Miro and DALL-E, were tested, limiting generalizability to other AI-assisted brainstorming platforms. The focus on visual stimuli was intentional, and tool use was counterbalanced to reduce ordering effects. Nonetheless, future studies should include text-based tools such as ChatGPT or hybrid platforms such as Canva, which may affect creativity differently depending on the task.

Snowball sampling led to a relatively homogeneous sample, particularly in digital competence and openness to innovation. Although most participants were unfamiliar with the tools, their high technological readiness may have influenced engagement. Future studies should use more stratified sampling to include varied ages, professions, and digital skills.

A maturity gap between tools also emerged: DALL-E, being newer and visually rich, may have benefited from a novelty effect relative to Miro. While counterbalancing and tutorials were used to mitigate this, longitudinal research could explore how familiarity shapes creative output over time and across experience levels.

Each brainstorming session lasted 15 minutes to sustain engagement and reduce cognitive fatigue, but this may have constrained idea development. Future work should compare

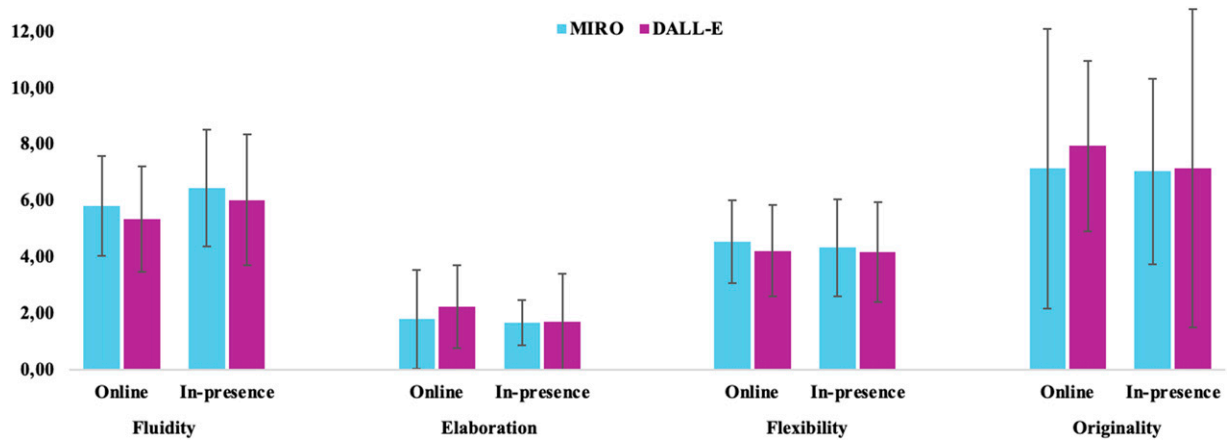


FIG. 6. Graphical visualization of creativity scores.

Downloaded by 37.159.0.213 from www.liebertpub.com at 06/18/25. For personal use only.

different session lengths under similar cognitive conditions to identify optimal durations.

Although many findings were statistically significant, effect sizes were often modest (e.g., $\eta^2p < 0.07$), potentially limiting practical implications. Replication with larger and more diverse samples is needed to strengthen generalizability.

Finally, free-riding appeared more frequently among adults (f_r 51%) than young adults (f_r 49%), especially with DALL-E, suggesting some overreliance on AI. While engagement was monitored during sessions, future systems could implement real-time feedback, turn-taking prompts, or contribution tracking to foster active collaboration.

Despite these limitations, which may affect the external validity of the study, several measures were implemented to mitigate potential biases, and the study provided valuable initial insights into human–AI interaction. Further research is needed to validate these findings across different settings, populations, and creative domains.

Conclusions

This study explored the impact of visual GAI tools versus traditional digital whiteboards in online and in-person brainstorming sessions. DALL-E enhanced positive affect, flow, and user experience, especially online, where it also improved perceived efficiency. Miro was more effective in in-person settings, suggesting that tool-context alignment is key in collaborative creativity. Creativity outcomes were generally similar, but online sessions led to more elaborated ideas across platforms.

Interaction effects reinforced these findings: DALL-E was perceived as more efficient and immersive online, while Miro performed better in-presence. This highlights that the effectiveness of each tool depends on the specific environment.

These findings indicated that GAI tools can serve as valuable allies in collaborative creativity. Their application may be particularly useful in digital environments, offering new opportunities for innovation in different contexts. Furthermore, they highlighted the importance of selecting tools that align with the specific brainstorming context and provided a foundation for future studies on human–AI co-creation. In this light, the results offered preliminary support for the emergence of a hybrid creative context, where human cognition, group dynamics, and generative AI collaboratively could shape the creative process. Future research could compare individual and group creative productivity using GAI tools, potentially exploring the effectiveness of nominal groups versus teams,[85,86] with a mixed-method design study.

Acknowledgment

The authors wish to thank Alessandro Roventi, who contributed to data collection.

Authors' Contributions

K.G.: Conceptualization (supporting), first draft (lead), formal analysis (supporting), and final version (lead). E.D.S.: Revision (supporting). F.B.: Formal analysis (lead). A.P.: Revision (supporting). A.G.: Revision (supporting) and supervision (supporting). A.C.: Conceptualization (lead), rational (lead), supervision (lead), and revision (lead).

Author Disclosure Statement

The authors have no competing interests to declare that are relevant to the content of this article.

Funding Information

The present work was supported by the Italian Ministry of University and Research under the program "PNRR per i dottorati industriali e innovativi che rispondono ai fabbisogni delle imprese" DM 117/2023 and by the Grants PRIN 2022 PNRR P2022PXAZW funded by European Union NextGeneration EU.

References

1. Winer RP. Creativity and Beyond: Cultures, Values, and Change. State University of New York press; n.d.
2. Davids K, Munck BD. Innovation and Creativity in Late Medieval and Early Modern European Cities. Routledge; 2016.
3. Montuori A, Purser RE. Deconstructing the lone genius myth: Toward a contextual view of creativity. J Humanist Psychol 1995;35(3):69–112; doi: 10.1177/00221678950353005
4. Eisler R, Donnelly G, Montuori A. Creativity, society, and gender: Contextualizing and redefining creativity. IJPS 2016;3(2); doi: 10.24926/IJPS.V3I2.130
5. Sawyer K. Group Genius: The Creative Power of Collaboration. Porchlight Book Company; 2007. (Basic Book. ed).
6. Negus K, Pickering M. Creativity, Communication and Cultural Value. Sage; 2004.
7. Guilford JP. The Nature of Human Intelligence. McGraw--Hill: New York; 1967.
8. Runco M, Albert R. Creativity research: A historical view. In: The Cambridge Handbook of Creativity. (Kaufman J, Stenberg R. eds) Cambridge University Press; 2010; pp. 3–19.
9. Torrance EP. Torrance Tests of Creative Thinking. Educ Psychol Meas; 1966; doi: 10.1037/T05532-000
10. Csikszentmihalyi M. Society, culture, and person: A systems view of creativity. In: The Nature of Creativity: Contemporary Psychological Perspectives. Cambridge University Press; 1988; pp. 325–339.
11. Glăveanu V. Creativity as a sociocultural act. J Creative Behavior 2015;49(3):165–180; doi: 10.1002/jocb.94
12. Hutchins E. The distributed cognition perspective on human interaction. Roots of Human Sociality: Culture, Cognition and Interaction. Routledge; 2020;375–398; doi: 10.4324/9781003135517-19/DISTRIBUTED-COGNITION-PERSPECTIVE-HUMAN-INTERACTION-EDWIN-HUTCHINS
13. Sawyer R, Henriksen D. Explaining Creativity: The Science of Human Innovation. Oxford University Press; 2024.
14. Lubart T, Esposito D, Gubenko A, et al. Creativity in humans, robots, humbots. Creativity 2021;8(1):23–37; doi: 10.2478/ctra-2021-0003
15. Huang Q, Jansen A, Lee J, et al. MuLan: A joint embedding of music audio and natural language. arXiv 2022.
16. Wang C, Chen S, Wu Y, et al. Neural codec language models are zero-shot text to speech synthesizers. arXiv 2023.
17. Dwivedi YK, Hughes L, Ismagilova E, et al. Artificial intelligence (AI): Multidisciplinary perspectives on emerging challenges, opportunities, and agenda for research, practice and policy. Int J Inf Manage 2021;57:101994; doi: 10.1016/j.ijinfomgt.2019.08.002
18. Joosten J, Bilgram V, Hahn A, et al. Comparing the ideation quality of humans with generative artificial intelligence.

Downloaded by 37.159.0.213 from www.liebertpub.com at 06/18/25. For personal use only.

IEEE Eng Manag Rev 2024;52(2):153–164; doi: 10.1109/EMR.2024.3353338
19. Hubert KF, Awa KN, Zabelina DL. The current state of artificial intelligence generative language models is more creative than humans on divergent thinking tasks. Sci Rep 2024; 14(1):3440; doi: 10.1038/s41598-024-53303-w
20. Kaboodi M, Jiar YK. Cognitive and trait creativity in relation with academic achievement. Ijssh 2013:391–395; doi: 10.7763/ijssh.2012.v2.132
21. Dellermann D, Ebel P, Söllner M, et al. Hybrid intelligence. Bus Inf Syst Eng 2019;61(5):637–643; doi: 10.1007/s12599-019-00595-2
22. Lichtenthaler U. Substitute or synthesis: The interplay between human and artificial intelligence. Res Technol Management 2018;61(5):12–14; doi: 10.1080/08956308.2018.1495962
23. Krogh von G. ETH library artificial intelligence in organizations: New opportunities for phenomenon-based theorizing. Academy Manag Discov 2018;4(4): 404–409; doi: 10.3929/ethz-b-000320207
24. Beaty R, Silvia P, Nusbaum E, et al. The roles of associative and executive processes in creative cognition. Mem Cognit 2014;42(7):1186–1197; doi: 10.3758/s13421-014-0428-8
25. Beaty R, Seli P, Schacter D. Network neuroscience of creative cognition: Mapping cognitive mechanisms and individual differences in the creative brain. Curr Opin Behav Sci 2019;27:22–30; doi: 10.1016/j.cobeha.2018.08.013
26. Sunavsky A, Poppenk J. Neuroimaging predictors of creativity in healthy adults. Neuroimage 2020;206:116292; doi: 10.1016/j.neuroimage.2019.116292
27. Mayer R. Fifty Years of Creativity Research. 1999.
28. Lavrič F, Škraba A. Brainstorming will never be the same again—A human group supported by artificial intelligence. MAKE 2023;5(4):1282–1301; doi: 10.3390/make5040065
29. Sbai O, Elhoseiny M, Bordes A, et al. Design: Design inspiration from generative networks. In: Proceedings of the European Conference on Computer Vision (ECCV) 2018.
30. Muller M, Chilton LB, Kantosalo A, et al. GenAICHI: Generative AI and HCI. Conference on Human Factors in Computing Systems—Proceedings 2022; doi: 10.1145/3491101.3503719
31. Haase J, Hanel PHP. Artificial muses: Generative artificial intelligence chatbots have risen to human-level creativity. J Creativity 2023;33(3):100066; doi: 10.1016/J.YJOC.2023.100066
32. Ochse R. Before the Gates of Excellence: The Determinants of Creative Genius. Cambridge University Press; 1990.
33. Simonton D. Scientific Genius: A Psychology of Science. Cambridge University Press; 1988.
34. Osborn. Applied Imagination: Principles and Procedures of Creative Problem Solving. Charles Scribner's Sons: New York, New York; 1953.
35. Stein M. Stimulating Creativity. Academic Press: New York; 1974.
36. Gero KI, Liu V, Chilton LB. Sparks: Inspiration for science writing using language models. Proceedings of the 2022 ACM Designing Interactive Systems Conference: Digital Wellbeing 2022;1002–1019; doi: 10.1145/3532106.3533533
37. Dang H, Mecke L, Lehmann F, et al. How to prompt? Opportunities and challenges of zero- and few-shot learning for human-AI interaction in creative applications of generative models. arXiv 2022; doi: 10.48550/arXiv.2209.01390
38. Wen M, Baym N, Tamuz O, et al. OMG UR Funny! Computer-Aided Humor with an Application to Chat. ICCC; 2015; pp. 86–93.
39. Davis N, Hsiao CP, Popova Y, et al. An Enactive Model of Creativity for Computational Collaboration and Co-Creation. Springer Series on Cultural Computing; 2015;109–133; doi: 10.1007/978-1-4471-6681-8_7
40. Croply AJ. Definition of Creativity. Encyclopedia of creativity; 1999;1:511–524.
41. May R. The nature of creativity. In: Etc. Cambridge University Press; 1959; pp. 261–276.
42. Runco MA. AI can only produce artificial creativity. J Creat 2023;33(3):100063; doi: 10.1016/j.yjoc.2023.100063
43. Vinchon F, Lubart T, Bartolotta S, et al. Artificial intelligence & creativity: A manifesto for collaboration. J Creative Behavior 2023;57(4):472–484; doi: 10.1002/jocb.597
44. Wu Z, Ji D, Yu K, et al. AI creativity and the human-AI co-creation model. In: Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics). Springer Science and Business Media Deutschland GmbH; 2021; pp. 171–190; doi: 10.1007/978-3-030-78462-1_13
45. Shaer O, Cooper A, Mokryn O, et al. AI-augmented brainwriting: Investigating the use of LLMs in group ideation. Conference on Human Factors in Computing Systems—Proceedings. 2024;17; doi: 10.1145/3613904.3642414
46. Urban M, Děchtěrenko F, Lukavský J, et al. ChatGPT improves creative problem-solving performance in university students: An experimental study. Comput Educ 2024; 215:105031; doi: 10.1016/J.COMPEDU.2024.105031
47. Dennis A, Valacich J. Computer brainstorms: More heads are better than one. J Appl Psychol 1993;78(4):531–537.
48. Siemon D, Eckardt L, Robra-Bissantz S. Tracking down the negative group creativity effects with the help of an artificial intelligence-like support system. In: In 2015 48th Hawaii International Conference on System Sciences (pp. 236–243). IEEE. 2015.
49. von Krogh G. Artificial intelligence in organizations: New opportunities for phenomenon-based theorizing. Amd 2018; 4(4):404–409; doi: 10.5465/AMD.2018.0084
50. Gan J, Dang Q, Western B, et al. AI-mediated group ideation. In: ECAADe 41. ECAADe; 2015; pp. 389–396.
51. Simeone L. Pushing divergence and promoting convergence in a speculative design process: Considerations on the role of AI as a co-creation partner. 2022; doi: 10.21606/drs.2022.197
52. Suh MM, Youngblom E. AI as social glue: Uncovering the roles of deep generative AI during social music composition. In: Conference on Human Factors in Computing Systems—Proceedings Association for Computing Machinery; 2021; doi: 10.1145/3411764.3445219
53. Petridis S, Diakopoulos N, Crowston K, et al. AngleKindling: Supporting journalistic angle ideation with large language models. In: Conference on Human Factors in Computing Systems—Proceedings Association for Computing Machinery; 2023; doi: 10.1145/3544548.3580907
54. Diehl M, Stroebe W. Productivity loss in brainstorming groups: Toward the solution of a riddle. J Pers Soc Psychol 1987;53(3):497–509.
55. Memmert L, Tavanapour N. Towards Human-AI-Collaboration in Brainstorming: Empirical Insights into the Perception of Working with a Generative AI. 2023.

Downloaded by 37.159.0.213 from www.liebertpub.com at 06/18/25. For personal use only.

56. Pinsonneault A, Barki H, Gallupe RB, et al. Electronic brainstorming: The illusion of productivity. Information Systems Res 1999;10(2):110–133; doi: 10.1287/ISRE.10.2.110
57. Jaco AA, Buisine S, Barrè J, et al. Trains of thought on the tabletop: Visualizing association of ideas improves creativity. Pers Ubiquit Comput 2014;18(5):1159–1167; doi: 10.1007/s00779-013-0726-3
58. Gumienny R, Meinel C, Gericke L, et al. Tele-board: Enabling efficient collaboration in digital design spaces across time and distance. Design Thinking 2011;147–164; doi: 10.1007/978-3-642-13757-0_9
59. Parasuraman A, Colby CL. An updated and streamlined technology readiness index: TRI 2.0. J Serv Res 2015;18(1):59–74; doi: 10.1177/1094670514539730
60. Terracciano A, McCrae RR, Costa PTJ. Factorial and construct validity of the Italian Positive and Negative Affect Schedule (PANAS). European J Psychological Assessment 2003;19(2):131–141.
61. Jackson SA, Marsh HW. Development and validation of a scale to measure optimal experience: The flow state scale. J Sport Exerc Psychol 1996;18(1):17–35; doi: 10.1123/JSEP.18.1.17
62. Laugwitz B, Held T, Schrepp M. Construction and evaluation of a User Experience Questionnaire. In: HCI and Usability for Education and Work: 4th Symposium of the Workgroup Human-Computer Interaction and Usability Engineering of the Austrian Computer Society. Springer Berlin Heidelberg: Graz, Austria; 2008; pp. 63–76; doi: 10.1007/978-3-540-89350-9_6
63. Schindler I, Hosoya G, Menninghaus W, et al. Measuring aesthetic emotions: A review of the literature and a new assessment tool. PLoS One 2017;12(6):e0178899; doi: 10.1371/journal.pone.0178899
64. Paulus PB, Kenworthy JB. Effective brainstorming. In: The Oxford Handbook of Group Creativity and Innovation. Oxford University Press; 2019; doi: 10.3389/FPSYG.2019.00699
65. Liikkanen LA, Björklund TA, Hämäläinen MM, et al. Time Constraints in Design Idea Generation. DS 58-9: Proceedings of ICED 09, the 17th International Conference on Engineering Design, Vol 9, Human Behavior in Design: Palo Alto, CA, USA, 24-27082009. 2009;81–90.
66. Menninghaus W, Wagner V, Hanich J, et al. The distancing-embracing model of the enjoyment of negative emotions in art reception. Behav Brain Sci 2017;40:e347; doi: 10.1017/S0140525X17000309
67. Nemeth CJ, Nemeth-Brown B. Better than individuals? The potential benefits of dissent and diversity for group creativity. In: Group Creativity: Innovation through Collaboration. Oxford University Press; 2003; doi: 10.1093/ACPROF:OSO/9780195147308.003.0004
68. Berlyne DE. Conflict, Arousal, and Curiosity. McGraw-Hill; 2006; doi: 10.1037/11164-000
69. Van der Heijden H. Factors influencing the usage of websites: The case of a generic portal in The Netherlands. Information & Management 2023.
70. Paulus PB, Nijstad BA. Group creativity: An introduction. In: Group Creativity: Innovation through Collaboration. Oxford University Press; 2003; doi: 10.1093/ACPROF:OSO/9780195147308.003.0001
71. Baas M, De Dreu CKW, Nijstad BA. A meta-analysis of 25 years of mood-creativity research: Hedonic tone, activation, or regulatory focus? Psychol Bull 2008;134(6):779–806; doi: 10.1037/A0012815
72. Fredrickson B, Branigan C. Positive emotions broaden the scope of attention and thought-action repertoires. Cogn Emot 2005;19(3):313–332; doi: 10.1080/02699930441000238
73. Landhäußer A, Keller J. Flow and its affective, cognitive, and performance-related consequences. In: Adv Flow Research. (Engeser S. ed) Springer: New York; 2012; pp. 65–85; doi: 10.1007/978-1-4614-2359-1_4
74. Rogatko TP. The influence of flow on positive affect in college students. J Happiness Stud 2009;10(2):133–148; doi: 10.1007/S10902-007-9069-Y
75. Zhang X, Pi Z, Li C, et al. Intrinsic motivation enhances online group creativity via promoting members' effort, not interaction. Brit J Educational Tech 2021;52(2):606–618; doi: 10.1111/BJET.13045
76. Kharkhurin AV. Creativity.4in1: Four-criterion construct of creativity. Creat Res J 2014;26(3):338–352; doi: 10.1080/10400419.2014.929424
77. Runco MA. Authentic creativity: Mechanisms, definitions, and empirical efforts. In: The Nature of Human Creativity. Sternberg RJ. & Kaufman JC. (Eds.), Cambridge University Press; 2018; pp. 246–263.
78. Gaggioli A, Mazzoni E, Benvenuti M, et al. Networked flow in creative collaboration: A mixed method study. Creat Res J 2020;32(1):41–54; doi: 10.1080/10400419.2020.1712160
79. Amabile TM, Ann Collins M, Conti R, et al. Other social and environmental influences. In: Creativity in Context. Routledge; 1983; pp. 159–185; doi: 10.4324/9780429501234-8/SOCIAL-ENVIRONMENTAL-INFLUENCES-TERESA-AMABILE-TERESA-AMABILE-MARY-ANN-COLLINS-REGINA-CONTI-ELISE-PHILLIPS-MARTHA-PICARIELLO-JOHN-RUSCIO-DEAN-WHITNEY
80. Diehl M, Stroebe W. Productivity loss in idea-generating groups: Tracking down the blocking effect. J Pers Soc Psychol 1991;61(3):392–403.
81. Wever M, Engelen J. Online vs Offline Brainstorming. The Influence of Feeling Heard on Satisfaction and Engagement in Ideation 2023.
82. Michinov N, Primois C. Improving productivity and creativity in online groups through social comparison process: New evidence for asynchronous electronic brainstorming. Comput Human Behav 2005;21(1):11–28; doi: 10.1016/J.CHB.2004.02.004
83. Dennis AR, Williams ML. Electronic brainstorming: Theory, research, and future directions. In: Group Creativity: Innovation through Collaboration. Oxford University Press; 2003; doi: 10.1093/ACPROF:OSO/9780195147308.003.0008

Address correspondence to:
Katusha Gerardini
Research Center in Communication Psychology (PsiCom)
Department of Psychology
Catholic University of the Sacred Heart
Largo Fra Agostino Gemelli
20123 Milan
Italy

E-mail: katusha.gerardini@unicatt.it

Downloaded by 37.159.0.213 from www.liebertpub.com at 06/18/25. For personal use only.